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1-Phase Full Wave Fully Controlled and Semi Controlled Rectifier

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Comprehensive Class Notes with Complete Derivations

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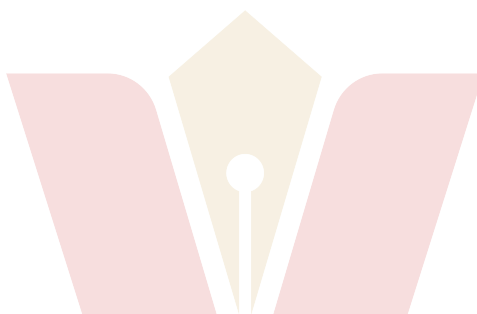
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1 Module Overview & Lecture Structure

The main aim of the present note is to derive, step by step, all performance expressions of the single-phase full wave phase controlled rectifier. Two circuits are considered, say the fully controlled converter (full converter) and the semi controlled converter (semi converter). Four load types are analysed, say R, RL, RE and RLE, each without and with free wheeling diode (FD). The notation is the same as that of the half wave note.

Definition 1.1 (Module Philosophy): *A full wave converter produces two output pulses in every supply cycle. The output quantities therefore repeat after every π radian. Every result in this note follows from two operations only: (i) writing KVL for the conducting interval and (ii) integrating over one output period π .*

1.1 Lecture Learning Structure

Table 1: 1-Phase Full Wave Controlled Rectifier - Lecture Structure

Lecture	Topic	Duration	Learning Activities
1	Circuits, angles and performance definitions	50 min	Waveform sketching
2	Full converter with R load	50 min	Derivation on board
3	Full converter with RL load without FD	50 min	Continuous and discontinuous modes
4	Full converter with RL load with FD	50 min	Mode wise analysis
5	Full converter with RE load	50 min	Battery charger problem
6	Full converter with RLE load	50 min	Rectification and inversion
7	Semi converter with all loads	50 min	Device current comparison
8	Source inductance and overlap angle	50 min	Numerical workshop
9	Numerical determination of β by NR method	50 min	Iteration by calculator

2 Circuit Description and Basic Definitions

2.1 Circuits Considered

Full converter (bridge): four thyristors form a bridge. T1 and T2 conduct together in the positive half cycle. T3 and T4 conduct together in the negative half cycle. T1, T2 are fired at $\omega t = \alpha$ and T3, T4 at $\omega t = \pi + \alpha$.

Semi converter (bridge): two thyristors and two diodes form the bridge. Two arrangements are in use:

- **Symmetrical:** T1 and T2 have common cathode, D1 and D2 have common anode. Each leg has one thyristor and one diode
- **Asymmetrical:** T1 and T2 are connected in series in one leg, with their junction at one AC terminal. D1 and D2 are connected in series in the other leg, with their junction at the other AC terminal. The two diodes in series are directly across the load and give a free wheeling path

Free wheeling diode: for the FD cases, a diode FD is connected across the load with its cathode at the positive load terminal.

Mid-point (centre tap) converter: two thyristors with a centre tapped transformer give the same output expressions as the bridge, if V_m is the peak voltage of each half of the secondary. Only the PIV differs.

Source voltage:

$$v_s = V_m \sin \omega t, \quad V_m = \sqrt{2} V_s \quad (1)$$

Assumptions:

- Thyristors and diodes are ideal
- Source is ideal, except in Section 9 where source inductance L_s is included
- Steady state operation. Output quantities have period π and input quantities have period 2π in ωt

2.2 Angle Definitions

Definition 2.1 (Angles Used in This Note):

- α = firing (delay) angle, measured from the zero crossing of v_s
- β = extinction angle, the value of ωt at which load current falls to zero
- $\gamma = \beta - \alpha$ = conduction angle of load current in each half cycle
- $\theta = \tan^{-1} \left(\frac{\omega L}{R} \right)$ = load impedance angle
- μ = overlap (commutation) angle due to source inductance L_s
- $\delta = \sin^{-1} \left(\frac{E}{V_m} \right)$ = angle at which v_s equals the load EMF E

Load impedance relations:

$$Z = \sqrt{R^2 + (\omega L)^2} \quad (2)$$

$$R = Z \cos \theta, \quad \omega L = Z \sin \theta, \quad \frac{R}{\omega L} = \frac{1}{\tan \theta} \quad (3)$$

Conduction modes:

- **Discontinuous conduction:** $\beta < \pi + \alpha$, i.e. $\gamma < \pi$. Load current is zero for some part of each half cycle
- **Continuous conduction:** load current never falls to zero. Each thyristor pair of the full converter conducts for π

2.3 Performance Parameter Definitions

Since the output waveform repeats after π :

Average (DC) values:

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} v_o d(\omega t), \quad I_o = \frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} i_o d(\omega t) \quad (4)$$

RMS values:

$$V_{or} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} v_o^2 d(\omega t)}, \quad I_{or} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} i_o^2 d(\omega t)} \quad (5)$$

Form factor and ripple factor:

$$FF = \frac{V_{or}}{V_o} \quad (6)$$

$$V_{or}^2 = V_o^2 + V_{ac}^2 \quad (7)$$

$$RF = \frac{V_{ac}}{V_o} = \sqrt{FF^2 - 1} \quad (8)$$

Input (source) current: let i_T be the current of the conducting thyristor pair. In the positive half cycle $i_s = +i_T$ and in the negative half cycle $i_s = -i_T$. During free wheeling $i_T = 0$ and $i_s = 0$. The source current is an alternating wave with half wave symmetry. Therefore:

$$I_{s,avg} = \frac{1}{2\pi} \int_0^{2\pi} i_s d(\omega t) = 0 \quad (9)$$

The source current has no DC component. This is a major advantage over the half wave circuit, since the supply transformer is not driven into saturation. The useful input quantity is the RMS value:

$$I_{s,rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} i_s^2 d(\omega t)} \quad (10)$$

Without FD (full converter), $|i_s| = i_o$ at every instant. Hence $I_{s,rms} = I_{or}$ and the half cycle average of i_s is I_o .

Input power factor:

$$pf = \frac{\text{Power delivered to load}}{V_s I_{s,rms}} \quad (11)$$

Peak inverse voltage (PIV): the maximum reverse voltage across a device in OFF state.

3 Full Converter with R Load

3.1 R Load without FD

3.1.1 Operation

- $\alpha \leq \omega t \leq \pi$: T1, T2 conduct. $v_o = v_s$, $i_o = v_s/R$, $i_s = i_o$
- $\pi < \omega t < \pi + \alpha$: current is zero at $\omega t = \pi$ and all devices are OFF. $v_o = 0$
- $\pi + \alpha \leq \omega t \leq 2\pi$: T3, T4 conduct. $v_o = -v_s > 0$, $i_s = -i_o$

Hence $\beta = \pi$ and $\gamma = \pi - \alpha$ in each half cycle.

3.1.2 Average Output Voltage

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d(\omega t) \quad (12)$$

$$= \frac{V_m}{\pi} \left[-\cos \omega t \right]_{\alpha}^{\pi} \quad (13)$$

$$V_o = \frac{V_m}{\pi} (1 + \cos \alpha) \quad (14)$$

This is twice the half wave value. At $\alpha = 0$, $V_o = 2V_m/\pi$.

3.1.3 RMS Output Voltage

$$V_{or}^2 = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m^2 \sin^2 \omega t d(\omega t) \quad (15)$$

$$= \frac{V_m^2}{2\pi} \left[\omega t - \frac{\sin 2\omega t}{2} \right]_{\alpha}^{\pi} \quad (16)$$

$$= \frac{V_m^2}{2\pi} \left[(\pi - \alpha) + \frac{\sin 2\alpha}{2} \right] \quad (17)$$

$$V_{or} = \frac{V_m}{\sqrt{2\pi}} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha} = V_s \sqrt{\frac{1}{\pi} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]} \quad (18)$$

At $\alpha = 0$, $V_{or} = V_s$.

3.1.4 Output, Input and Device Currents

$$I_o = \frac{V_o}{R} = \frac{V_m}{\pi R} (1 + \cos \alpha) \quad (19)$$

$$I_{or} = \frac{V_{or}}{R} \quad (20)$$

$$I_{s,avg} = 0, \quad I_{s,rms} = I_{or} \quad (21)$$

Each thyristor carries one current pulse out of two in every supply cycle:

$$I_{T,avg} = \frac{I_o}{2}, \quad I_{T,rms} = \frac{I_{or}}{\sqrt{2}} \quad (22)$$

3.1.5 Form Factor and Ripple Factor

$$FF = \frac{V_{or}}{V_o} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha}}{1 + \cos \alpha} \quad (23)$$

$$RF = \sqrt{FF^2 - 1} \quad (24)$$

At $\alpha = 0$: $FF = \frac{\pi}{2\sqrt{2}} = 1.111$ and $RF = 0.483$.

3.1.6 PIV and Other Results

PIV (bridge): when T3, T4 conduct, T1 is connected directly across the source in reverse direction. The reverse voltage is $|v_s|$.

$$PIV = V_m \quad (\text{bridge}), \quad PIV = 2V_m \quad (\text{mid-point}) \quad (25)$$

Load power and input power factor:

$$P = \frac{V_{or}^2}{R} \quad (26)$$

$$pf = \frac{V_{or}^2/R}{V_s \cdot V_{or}/R} = \frac{V_{or}}{V_s} = \sqrt{\frac{1}{\pi} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]} \quad (27)$$

At $\alpha = 0$, $pf = 1$.

Rectification efficiency:

$$\eta = \frac{V_o I_o}{V_{or} I_{or}} = \frac{1}{FF^2} \quad (28)$$

At $\alpha = 0$, $\eta = 8/\pi^2 = 81.06\%$.

3.2 R Load with FD

With R load, $v_o = Ri_o \geq 0$ at every instant. The FD is never forward biased. All expressions of R load without FD remain valid.

Example 3.1 (Full Converter with R Load):

A 1-phase full converter feeds $R = 10 \Omega$ from 230 V, 50 Hz supply at $\alpha = 60^\circ$.

Peak voltage:

$$V_m = \sqrt{2} \times 230 = 325.27 \text{ V}, \quad \frac{V_m}{\pi} = 103.54 \text{ V} \quad (29)$$

Average output voltage and current:

$$V_o = 103.54 (1 + \cos 60^\circ) = 155.31 \text{ V} \quad (30)$$

$$I_o = \frac{155.31}{10} = 15.53 \text{ A} \quad (31)$$

RMS output voltage and current:

$$V_{or} = 230 \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{3} + \frac{\sin 120^\circ}{2} \right)} = 230 \sqrt{\frac{2.0944 + 0.4330}{3.1416}} = 230 \times 0.8969 = 206.30 \text{ V} \quad (32)$$

$$I_{or} = 20.63 \text{ A} \quad (33)$$

Form factor and ripple factor:

$$\text{FF} = \frac{206.30}{155.31} = 1.328 \quad (34)$$

$$\text{RF} = \sqrt{1.328^2 - 1} = 0.874 \quad (35)$$

Input and device quantities:

$$I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = 20.63 \text{ A} \quad (36)$$

$$I_{T,\text{avg}} = \frac{15.53}{2} = 7.77 \text{ A}, \quad I_{T,\text{rms}} = \frac{20.63}{\sqrt{2}} = 14.59 \text{ A} \quad (37)$$

$$\text{pf} = \frac{206.30}{230} = 0.897 \text{ (lag)} \quad (38)$$

$$\text{PIV} = 325.27 \text{ V} \quad (39)$$

Comparison: the half wave circuit gives $\text{FF} = 1.879$ and $\text{RF} = 1.590$ for the same data.

4 Full Converter with RL Load without FD

4.1 Operation

The pair T1, T2 is fired at $\omega t = \alpha$. Due to L , the current continues beyond $\omega t = \pi$ and $v_o = v_s$ becomes negative. Two cases arise:

- **Discontinuous conduction:** the current falls to zero at $\omega t = \beta < \pi + \alpha$. All devices are OFF and $v_o = 0$ till T3, T4 are fired
- **Continuous conduction:** the current is still flowing when T3, T4 are fired at $\omega t = \pi + \alpha$. T1, T2 are then turned OFF by line commutation

4.2 Discontinuous Conduction

4.2.1 Load Current Expression

Step 1: KVL during conduction

$$L \frac{di_o}{dt} + R i_o = V_m \sin \omega t, \quad \alpha \leq \omega t \leq \beta \quad (40)$$

Step 2: complete solution (forced component plus natural component)

$$i_o = \frac{V_m}{Z} \sin(\omega t - \theta) + A e^{-\frac{\omega t - \alpha}{\tan \theta}} \quad (41)$$

Step 3: initial condition $i_o(\alpha) = 0$ gives $A = -\frac{V_m}{Z} \sin(\alpha - \theta)$.

Step 4: final expression

$$i_o(\omega t) = \frac{V_m}{Z} \left[\sin(\omega t - \theta) - \sin(\alpha - \theta) e^{-\frac{\omega t - \alpha}{\tan \theta}} \right], \quad \alpha \leq \omega t \leq \beta \quad (42)$$

Step 5: extinction angle from $i_o(\beta) = 0$:

$$\sin(\beta - \theta) = \sin(\alpha - \theta) e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (43)$$

This is the same equation as that of the half wave circuit. It is solved numerically (Section 10).

4.2.2 Average and RMS Output Voltage

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\beta} V_m \sin \omega t d(\omega t) = \frac{V_m}{\pi} (\cos \alpha - \cos \beta) \quad (44)$$

$$V_{or}^2 = \frac{1}{\pi} \int_{\alpha}^{\beta} V_m^2 \sin^2 \omega t d(\omega t) = \frac{V_m^2}{2\pi} \left[(\beta - \alpha) - \frac{1}{2} (\sin 2\beta - \sin 2\alpha) \right] \quad (45)$$

$$V_{or} = V_s \sqrt{\frac{1}{\pi} \left[\gamma - \frac{1}{2} (\sin 2\beta - \sin 2\alpha) \right]} \quad (46)$$

4.2.3 Average and RMS Output Current

The average inductor voltage over one period is zero, since $i_o(\alpha) = i_o(\beta) = 0$:

$$I_o = \frac{V_o}{R} = \frac{V_m}{\pi R} (\cos \alpha - \cos \beta) \quad (47)$$

For the RMS current let $K = \sin(\alpha - \theta)$. Squaring (42) and integrating:

$$J_1 = \int_{\alpha}^{\beta} \sin^2(\omega t - \theta) d(\omega t) = \frac{\gamma}{2} - \frac{1}{4} [\sin 2(\beta - \theta) - \sin 2(\alpha - \theta)] \quad (48)$$

$$J_2 = \int_{\alpha}^{\beta} K^2 e^{-\frac{2(\omega t - \alpha)}{\tan \theta}} d(\omega t) = \frac{K^2 \tan \theta}{2} (1 - e^{-\frac{2\gamma}{\tan \theta}}) \quad (49)$$

$$J_3 = \int_{\alpha}^{\beta} \sin(\omega t - \theta) e^{-\frac{\omega t - \alpha}{\tan \theta}} d(\omega t) = \sin \theta \left(\sin \alpha - e^{-\frac{\gamma}{\tan \theta}} \sin \beta \right) \quad (50)$$

$$I_{or} = \frac{V_m}{Z} \sqrt{\frac{1}{\pi} (J_1 + J_2 - 2K J_3)} \quad (51)$$

Check: for $L = 0$, $\theta = 0$ and $\beta = \pi$. Then $J_2 = J_3 = 0$, $J_1 = \frac{1}{2}(\pi - \alpha) + \frac{1}{4} \sin 2\alpha$ and the R load result is obtained.

4.2.4 Input Current, FF, RF and PIV

$$I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = I_{or} \quad (52)$$

$$I_{T,\text{avg}} = \frac{I_o}{2}, \quad I_{T,\text{rms}} = \frac{I_{or}}{\sqrt{2}} \quad (53)$$

$$\text{FF} = \frac{V_{or}}{V_o} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sqrt{\gamma - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)}}{\cos \alpha - \cos \beta}, \quad \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (54)$$

$$\text{PIV} = V_m \quad (55)$$

$$\text{pf} = \frac{I_{or}^2 R}{V_s I_{or}} = \frac{I_{or} R}{V_s} \quad (56)$$

4.3 Continuous Conduction

4.3.1 Average Output Voltage

Each pair conducts from α to $\pi + \alpha$ and $v_o = v_s$ throughout:

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} V_m \sin \omega t d(\omega t) \quad (57)$$

$$= \frac{V_m}{\pi} [\cos \alpha - \cos(\pi + \alpha)] \quad (58)$$

$$V_o = \frac{2V_m}{\pi} \cos \alpha \quad (59)$$

V_o is positive for $\alpha < \pi/2$, zero at $\alpha = \pi/2$ and negative for $\alpha > \pi/2$.

4.3.2 RMS Output Voltage

$$V_{or}^2 = \frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} V_m^2 \sin^2 \omega t d(\omega t) \quad (60)$$

$$= \frac{V_m^2}{2\pi} \left[\pi - \frac{1}{2} \{ \sin(2\pi + 2\alpha) - \sin 2\alpha \} \right] = \frac{V_m^2}{2} \quad (61)$$

$$V_{or} = \frac{V_m}{\sqrt{2}} = V_s \quad (62)$$

The RMS output voltage is independent of α , since the load is always connected to the source.

4.3.3 Form Factor and Ripple Factor

$$\text{FF} = \frac{V_s}{(2V_m/\pi) \cos \alpha} = \frac{\pi}{2\sqrt{2} \cos \alpha} \quad (63)$$

$$\text{RF} = \sqrt{\frac{\pi^2}{8 \cos^2 \alpha} - 1} \quad (64)$$

4.3.4 Load Current Expression

Let $I_\alpha = i_o(\alpha)$. The solution of the KVL equation is:

$$i_o(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) + A e^{-\frac{\omega t - \alpha}{\tan \theta}}, \quad A = I_\alpha - \frac{V_m}{Z} \sin(\alpha - \theta) \quad (65)$$

Steady state condition: $i_o(\pi + \alpha) = I_\alpha$. Using $\sin(\pi + \alpha - \theta) = -\sin(\alpha - \theta)$:

$$I_\alpha = -\frac{V_m}{Z} \sin(\alpha - \theta) + A e^{-\frac{\pi}{\tan \theta}} \quad (66)$$

$$A - I_\alpha = -\frac{V_m}{Z} \sin(\alpha - \theta) \quad (\text{from the definition of } A) \quad (67)$$

Adding the two relations:

$$A \left(1 - e^{-\frac{\pi}{\tan \theta}}\right) = -\frac{2V_m}{Z} \sin(\alpha - \theta) \Rightarrow A = \frac{2V_m}{Z} \cdot \frac{\sin(\theta - \alpha)}{1 - e^{-\frac{\pi}{\tan \theta}}} \quad (68)$$

$$i_o(\omega t) = \frac{V_m}{Z} \left[\sin(\omega t - \theta) + \frac{2 \sin(\theta - \alpha)}{1 - e^{-\frac{\pi}{\tan \theta}}} e^{-\frac{\omega t - \alpha}{\tan \theta}} \right], \quad \alpha \leq \omega t \leq \pi + \alpha \quad (69)$$

Current at the firing instant:

$$I_\alpha = \frac{V_m}{Z} \sin(\theta - \alpha) \cdot \frac{1 + e^{-\frac{\pi}{\tan \theta}}}{1 - e^{-\frac{\pi}{\tan \theta}}} = \frac{V_m}{Z} \sin(\theta - \alpha) \coth \left(\frac{\pi}{2 \tan \theta} \right) \quad (70)$$

Condition for continuous conduction: $I_\alpha \geq 0$ requires $\sin(\theta - \alpha) \geq 0$:

$$\alpha \leq \theta \quad (71)$$

For $\alpha > \theta$ the conduction is discontinuous. At $\alpha = \theta$ the current is just continuous and is a rectified sine wave.

4.3.5 Average Currents

$$I_o = \frac{V_o}{R} = \frac{2V_m}{\pi R} \cos \alpha \quad (72)$$

$$I_{T,\text{avg}} = \frac{I_o}{2}, \quad I_{T,\text{rms}} = \frac{I_{or}}{\sqrt{2}}, \quad I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = I_{or} \quad (73)$$

4.3.6 Highly Inductive Load (Ripple Free Current)

For $\omega L \gg R$ the load current is constant at I_o . The source current is a square wave of amplitude $\pm I_o$, shifted by α from the voltage.

$$I_{or} = I_o, \quad I_{s,\text{rms}} = I_o \quad (74)$$

$$I_{T,\text{avg}} = \frac{I_o}{2}, \quad I_{T,\text{rms}} = \frac{I_o}{\sqrt{2}} \quad (75)$$

Fundamental component of source current (Fourier series of a square wave):

$$I_{s1} = \frac{4I_o}{\pi\sqrt{2}} = \frac{2\sqrt{2}}{\pi} I_o = 0.9 I_o \quad (76)$$

Displacement factor, distortion factor and input power factor:

$$\text{DF} = \cos \alpha \quad (77)$$

$$\frac{I_{s1}}{I_{s,\text{rms}}} = \frac{2\sqrt{2}}{\pi} = 0.9 \quad (78)$$

$$\text{pf} = \frac{V_o I_o}{V_s I_{s,\text{rms}}} = \frac{2\sqrt{2}}{\pi} \cos \alpha = 0.9 \cos \alpha \quad (79)$$

Total harmonic distortion of source current:

$$\text{THD} = \sqrt{\left(\frac{I_{s,\text{rms}}}{I_{s1}}\right)^2 - 1} = \sqrt{\frac{\pi^2}{8} - 1} = 48.34\% \quad (80)$$

4.3.7 PIV and Turn-off Time

$$\text{PIV} = V_m \quad (\text{bridge}), \quad \text{PIV} = 2V_m \quad (\text{mid-point}) \quad (81)$$

T1 is reverse biased from $\omega t = \pi + \alpha$ to 2π . Circuit turn-off time is $t_c = (\pi - \alpha)/\omega$.

Example 4.1 (Full Converter with Highly Inductive Load):

A 1-phase full converter feeds a highly inductive load with $R = 5 \, \Omega$ from 230 V, 50 Hz supply at $\alpha = 45^\circ$.

Output voltage and current:

$$V_o = \frac{2 \times 325.27}{\pi} \cos 45^\circ = 207.07 \times 0.7071 = 146.42 \, \text{V} \quad (82)$$

$$I_o = \frac{146.42}{5} = 29.28 \, \text{A} \quad (83)$$

RMS output voltage, FF and RF:

$$V_{or} = 230 \, \text{V} \quad (84)$$

$$\text{FF} = \frac{230}{146.42} = 1.571, \quad \text{RF} = \sqrt{1.571^2 - 1} = 1.211 \quad (85)$$

Input and device currents:

$$I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = 29.28 \, \text{A} \quad (86)$$

$$I_{T,\text{avg}} = 14.64 \, \text{A}, \quad I_{T,\text{rms}} = \frac{29.28}{\sqrt{2}} = 20.71 \, \text{A} \quad (87)$$

Input power factor and PIV:

$$\text{pf} = 0.9 \times \cos 45^\circ = 0.637 \, (\text{lag}) \quad (88)$$

$$\text{PIV} = 325.27 \, \text{V} \quad (89)$$

5 Full Converter with RL Load with FD

5.1 Operation

At $\omega t = \pi$ the source voltage reverses. The FD becomes forward biased and takes over the load current. The conducting thyristor pair turns OFF. The negative part of v_o is removed.

- **Mode 1** ($\alpha \leq \omega t \leq \pi$): T1, T2 conduct, $v_o = v_s$, $i_s = i_o$
- **Mode 2** ($\pi < \omega t < \pi + \alpha$): FD conducts, $v_o = 0$, $i_s = 0$

5.2 Average and RMS Output Voltage

The output voltage waveform is the same as that of R load:

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t \, d(\omega t) = \frac{V_m}{\pi} (1 + \cos \alpha) \quad (90)$$

$$V_{or} = V_s \sqrt{\frac{1}{\pi} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]} \quad (91)$$

$$\text{FF} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha}}{1 + \cos \alpha}, \quad \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (92)$$

5.3 Load Current Expression

Let $I_\alpha = i_o(\alpha)$ and $I_\pi = i_o(\pi)$.

Mode 1:

$$i_o(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) + \left[I_\alpha - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\omega t - \alpha}{\tan \theta}} \quad (93)$$

$$I_\pi = \frac{V_m}{Z} \sin \theta + \left[I_\alpha - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\pi - \alpha}{\tan \theta}} \quad (94)$$

Mode 2: KVL is $L di_o/dt + R i_o = 0$.

$$i_o(\omega t) = I_\pi e^{-\frac{\omega t - \pi}{\tan \theta}} \quad (95)$$

Steady state condition: $i_o(\pi + \alpha) = I_\alpha$.

$$I_\alpha = I_\pi e^{-\frac{\alpha}{\tan \theta}} \quad (96)$$

Substituting (96) in (94):

$$I_\pi = \frac{V_m}{Z} \cdot \frac{\sin \theta - \sin(\alpha - \theta) e^{-\frac{\pi - \alpha}{\tan \theta}}}{1 - e^{-\frac{\pi}{\tan \theta}}} \quad (97)$$

With ideal devices the free wheeling current decays exponentially and does not reach zero in the finite interval α . Hence $I_\alpha > 0$ and the load current is mathematically continuous for RL load with FD. For small L/R or large α , I_α becomes negligible and the current is treated as discontinuous ($I_\alpha \approx 0$) in practice. The expressions of V_o and V_{or} are not affected.

5.4 Average Output Current

$$I_o = \frac{V_o}{R} = \frac{V_m}{\pi R} (1 + \cos \alpha) \quad (98)$$

5.5 Highly Inductive Load (Ripple Free Current)

Each thyristor conducts I_o for an angle $(\pi - \alpha)$ once in 2π . The FD conducts I_o for an angle α twice in 2π . The source current is $+I_o$ for $(\pi - \alpha)$, zero for α and $-I_o$ for $(\pi - \alpha)$.

$$I_{T,avg} = I_o \frac{\pi - \alpha}{2\pi}, \quad I_{T,rms} = I_o \sqrt{\frac{\pi - \alpha}{2\pi}} \quad (99)$$

$$I_{FD,avg} = I_o \frac{\alpha}{\pi}, \quad I_{FD,rms} = I_o \sqrt{\frac{\alpha}{\pi}} \quad (100)$$

$$I_{s,avg} = 0, \quad I_{s,rms} = I_o \sqrt{\frac{\pi - \alpha}{\pi}} \quad (101)$$

Fundamental component and displacement factor: the current pulse is centred at $\omega t = (\pi + \alpha)/2$, i.e. it lags the voltage peak by $\alpha/2$.

$$I_{s1} = \frac{2\sqrt{2}}{\pi} I_o \cos \frac{\alpha}{2}, \quad \text{DF} = \cos \frac{\alpha}{2} \quad (102)$$

Input power factor:

$$\text{pf} = \frac{V_o I_o}{V_s I_{s,rms}} = \frac{(V_m/\pi)(1 + \cos \alpha)}{V_s \sqrt{(\pi - \alpha)/\pi}} = \frac{\sqrt{2}(1 + \cos \alpha)}{\sqrt{\pi(\pi - \alpha)}} \quad (103)$$

5.6 PIV

- **Thyristor:** $PIV_T = V_m$ (bridge)
- **FD:** it blocks v_o during mode 1. $PIV_{FD} = V_m$ for $\alpha \leq \pi/2$ and $V_m \sin \alpha$ for $\alpha > \pi/2$

6 Full Converter with RE Load

6.1 RE Load without FD

6.1.1 Operation

A thyristor pair is forward biased only when $|v_s| > E$.

$$\delta = \sin^{-1} \left(\frac{E}{V_m} \right), \quad \delta \leq \alpha \leq \pi - \delta \quad (104)$$

In each half cycle the current flows from α to $\beta = \pi - \delta$. Hence $\gamma = \pi - \delta - \alpha$.

- $\alpha \leq \omega t \leq \pi - \delta$: T1, T2 conduct, $v_o = v_s$, $i_o = (V_m \sin \omega t - E)/R$
- Rest of the half cycle: all devices are OFF, $i_o = 0$, $v_o = E$

6.1.2 Average Output Current

$$I_o = \frac{1}{\pi R} \int_{\alpha}^{\pi-\delta} (V_m \sin \omega t - E) d(\omega t) \quad (105)$$

$$= \frac{1}{\pi R} \left[-V_m \cos \omega t - E \omega t \right]_{\alpha}^{\pi-\delta} \quad (106)$$

$$I_o = \frac{1}{\pi R} [V_m (\cos \alpha + \cos \delta) - E \gamma] \quad (107)$$

6.1.3 Average Output Voltage

The load voltage is v_s for an angle γ and E for the remaining angle $(\pi - \gamma) = (\alpha + \delta)$:

$$V_o = \frac{1}{\pi} \left[\int_{\alpha}^{\pi-\delta} V_m \sin \omega t d(\omega t) + E(\pi - \gamma) \right] \quad (108)$$

$$V_o = \frac{1}{\pi} [V_m (\cos \alpha + \cos \delta) + E(\alpha + \delta)] \quad (109)$$

Check: $V_o = E + R I_o$ gives the same expression.

6.1.4 RMS Output Current

$$I_{or}^2 = \frac{1}{\pi R^2} \int_{\alpha}^{\pi-\delta} (V_m^2 \sin^2 \omega t - 2V_m E \sin \omega t + E^2) d(\omega t) \quad (110)$$

Evaluating term by term, with $V_m^2/2 = V_s^2$ and $\sin 2(\pi - \delta) = -\sin 2\delta$:

$$\int_{\alpha}^{\pi-\delta} V_m^2 \sin^2 \omega t d(\omega t) = V_s^2 \left[\gamma + \frac{1}{2} (\sin 2\alpha + \sin 2\delta) \right] \quad (111)$$

$$\int_{\alpha}^{\pi-\delta} 2V_m E \sin \omega t d(\omega t) = 2V_m E (\cos \alpha + \cos \delta) \quad (112)$$

$$\int_{\alpha}^{\pi-\delta} E^2 d(\omega t) = E^2 \gamma \quad (113)$$

$$I_{or} = \sqrt{\frac{1}{\pi R^2} \left[(V_s^2 + E^2) \gamma + \frac{V_s^2}{2} (\sin 2\alpha + \sin 2\delta) - 2V_m E (\cos \alpha + \cos \delta) \right]} \quad (114)$$

6.1.5 RMS Output Voltage

$$V_{or} = \sqrt{\frac{1}{\pi} \left[V_s^2 \left\{ \gamma + \frac{1}{2} (\sin 2\alpha + \sin 2\delta) \right\} + E^2 (\alpha + \delta) \right]} \quad (115)$$

6.1.6 Input Current, FF, RF and PIV

$$I_{s,avg} = 0, \quad I_{s,rms} = I_{or} \quad (116)$$

$$I_{T,avg} = \frac{I_o}{2}, \quad I_{T,rms} = \frac{I_{or}}{\sqrt{2}} \quad (117)$$

$$FF = \frac{V_{or}}{V_o}, \quad RF = \sqrt{FF^2 - 1}, \quad RF_i = \sqrt{\left(\frac{I_{or}}{I_o}\right)^2 - 1} \quad (118)$$

PIV (bridge): when the other pair conducts, the OFF thyristor is directly across the source and sees $|v_s|$. When no device conducts, two OFF thyristors in series share $(v_s - E)$. The design value is:

$$PIV = V_m \quad (\text{bridge}) \quad (119)$$

Power relations:

$$P_E = E I_o, \quad P_R = I_{or}^2 R \quad (120)$$

$$pf = \frac{I_{or}^2 R + E I_o}{V_s I_{or}} \quad (121)$$

6.2 RE Load with FD

The load voltage is either $|v_s| > E$ or E . Thus $v_o > 0$ at every instant and the FD never conducts. All expressions of RE load without FD remain valid.

Example 6.1 (Full Converter with RE Load: Battery Charging):

A 120 V battery is charged through $R = 8 \Omega$ from 230 V, 50 Hz supply using a 1-phase full converter at $\alpha = 40^\circ$.

Angles:

$$\delta = \sin^{-1} \left(\frac{120}{325.27} \right) = 21.65^\circ = 0.3779 \text{ rad} \quad (122)$$

$$\beta = 180^\circ - 21.65^\circ = 158.35^\circ \quad (123)$$

$$\gamma = 158.35^\circ - 40^\circ = 118.35^\circ = 2.0656 \text{ rad} \quad (124)$$

Average charging current:

$$I_o = \frac{1}{\pi \times 8} [325.27 (0.7660 + 0.9295) - 120 \times 2.0656] \quad (125)$$

$$= \frac{551.50 - 247.87}{25.133} = 12.08 \text{ A} \quad (126)$$

RMS current:

$$I_{or}^2 = \frac{1}{\pi \times 64} [67300 \times 2.0656 + 26450 (0.9848 + 0.6858) - 78064.8 \times 1.6955] \quad (127)$$

$$= \frac{139014.9 + 44187.6 - 132358.9}{201.06} = 252.88 \quad (128)$$

$$I_{or} = 15.90 \text{ A} \quad (129)$$

Other results:

$$V_o = E + R I_o = 120 + 8 \times 12.08 = 216.64 \text{ V} \quad (130)$$

$$P_E = 120 \times 12.08 = 1449.6 \text{ W} \quad (131)$$

$$P_R = 252.88 \times 8 = 2023.0 \text{ W} \quad (132)$$

$$\text{pf} = \frac{1449.6 + 2023.0}{230 \times 15.90} = 0.950 \text{ (lag)} \quad (133)$$

$$I_{T,\text{avg}} = 6.04 \text{ A}, \quad I_{T,\text{rms}} = \frac{15.90}{\sqrt{2}} = 11.24 \text{ A} \quad (134)$$

Comparison: the charging current is twice that of the half wave circuit for the same data.

7 Full Converter with RLE Load

7.1 RLE Load without FD: Discontinuous Conduction

7.1.1 Load Current Expression

Step 1: KVL during conduction

$$L \frac{di_o}{dt} + R i_o + E = V_m \sin \omega t \quad (135)$$

Step 2: complete solution

$$i_o = \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} + A e^{-\frac{\omega t - \alpha}{\tan \theta}} \quad (136)$$

Step 3: initial condition $i_o(\alpha) = 0$ gives $A = \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta)$.

Step 4: final expression

$$i_o(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\omega t - \alpha}{\tan \theta}}, \quad \alpha \leq \omega t \leq \beta \quad (137)$$

Extinction angle: $i_o(\beta) = 0$ gives a transcendental equation, which is solved numerically (Section 10).

7.1.2 Average Output Voltage and Current

The load voltage is v_s for γ and E for $(\pi - \gamma)$:

$$V_o = \frac{1}{\pi} [V_m (\cos \alpha - \cos \beta) + E(\pi + \alpha - \beta)] \quad (138)$$

$$I_o = \frac{V_o - E}{R} = \frac{1}{\pi R} [V_m (\cos \alpha - \cos \beta) - E \gamma] \quad (139)$$

7.1.3 RMS Output Voltage and Current

$$V_{or} = \sqrt{\frac{1}{\pi} \left[V_s^2 \left\{ \gamma - \frac{1}{2} (\sin 2\beta - \sin 2\alpha) \right\} + E^2 (\pi - \gamma) \right]} \quad (140)$$

$$I_{or} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\beta} i_o^2 d(\omega t)} \quad (141)$$

with i_o from (137). The current integral is evaluated numerically.

7.2 RLE Load without FD: Continuous Conduction

7.2.1 Output Voltage and Current

The load is always connected to the source, so the voltage results of RL load hold:

$$V_o = \frac{2V_m}{\pi} \cos \alpha, \quad V_{or} = V_s \quad (142)$$

$$\text{FF} = \frac{\pi}{2\sqrt{2} \cos \alpha}, \quad \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (143)$$

$$I_o = \frac{V_o - E}{R} = \frac{1}{R} \left[\frac{2V_m}{\pi} \cos \alpha - E \right] \quad (144)$$

7.2.2 Load Current Expression

With $i_o(\alpha) = i_o(\pi + \alpha) = I_{\alpha}$, the constant A is found exactly as for RL load:

$$i_o(\omega t) = \frac{V_m}{Z} \left[\sin(\omega t - \theta) + \frac{2 \sin(\theta - \alpha)}{1 - e^{-\frac{\pi}{\tan \theta}}} e^{-\frac{\omega t - \alpha}{\tan \theta}} \right] - \frac{E}{R} \quad (145)$$

$$I_{\alpha} = \frac{V_m}{Z} \sin(\theta - \alpha) \coth \left(\frac{\pi}{2 \tan \theta} \right) - \frac{E}{R} \quad (146)$$

Condition for continuous conduction: $I_{\alpha} \geq 0$.

$$\frac{V_m}{Z} \sin(\theta - \alpha) \coth \left(\frac{\pi}{2 \tan \theta} \right) \geq \frac{E}{R} \quad (147)$$

7.2.3 Input and Device Currents

$$I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = I_{or}, \quad I_{T,\text{avg}} = \frac{I_o}{2}, \quad I_{T,\text{rms}} = \frac{I_{or}}{\sqrt{2}} \quad (148)$$

For ripple free current, $I_{or} = I_{s,\text{rms}} = I_o$ and $\text{pf} = 0.9 \cos \alpha$.

7.2.4 Inverter Operation

For $\alpha > \pi/2$, V_o is negative while I_o remains positive. Power $V_o I_o$ is negative, i.e. power flows from the DC side to the AC source. This is possible only if the load EMF is reversed ($E < 0$) and $|E| > |V_o|$. The full converter then works as a line commutated inverter.

7.2.5 PIV

$$\text{PIV} = V_m \quad (\text{bridge}), \quad \text{PIV} = 2V_m \quad (\text{mid-point}) \quad (149)$$

7.3 RLE Load with FD

7.3.1 Operation

- **Mode 1** ($\alpha \leq \omega t \leq \pi$): thyristor pair conducts, $v_o = v_s$
- **Mode 2** ($\pi < \omega t \leq \beta$): FD conducts, $v_o = 0$
- **Mode 3** ($\beta < \omega t < \pi + \alpha$): all devices are OFF, $i_o = 0$, $v_o = E$. This mode exists only for discontinuous conduction

7.3.2 Load Current Expression

Mode 1:

$$i_o(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} + \left[I_\alpha + \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\omega t - \alpha}{\tan \theta}} \quad (150)$$

$$I_\pi = \frac{V_m}{Z} \sin \theta - \frac{E}{R} + \left[I_\alpha + \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\pi - \alpha}{\tan \theta}} \quad (151)$$

Mode 2: KVL is $L di_o/dt + Ri_o + E = 0$.

$$i_o(\omega t) = -\frac{E}{R} + \left(I_\pi + \frac{E}{R} \right) e^{-\frac{\omega t - \pi}{\tan \theta}} \quad (152)$$

Extinction angle (discontinuous conduction, $I_\alpha = 0$):

$$\beta = \pi + \tan \theta \ln \left(1 + \frac{R I_\pi}{E} \right) \quad (153)$$

Conduction is discontinuous if $\beta < \pi + \alpha$. If the computed I_π is not positive, the current has become zero before $\omega t = \pi$. The FD then never conducts and β is found from (137).

7.3.3 Continuous Conduction

$$V_o = \frac{V_m}{\pi} (1 + \cos \alpha) \quad (154)$$

$$V_{or} = V_s \sqrt{\frac{1}{\pi} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]} \quad (155)$$

$$I_o = \frac{V_o - E}{R} = \frac{1}{R} \left[\frac{V_m}{\pi} (1 + \cos \alpha) - E \right] \quad (156)$$

For ripple free current the device and source currents are the same as those of RL load with FD:

$$I_{T,avg} = I_o \frac{\pi - \alpha}{2\pi}, \quad I_{FD,avg} = I_o \frac{\alpha}{\pi}, \quad I_{s,rms} = I_o \sqrt{\frac{\pi - \alpha}{\pi}} \quad (157)$$

7.3.4 Discontinuous Conduction

The load voltage is v_s for $(\pi - \alpha)$, zero for $(\beta - \pi)$ and E for $(\pi + \alpha - \beta)$:

$$V_o = \frac{1}{\pi} [V_m (1 + \cos \alpha) + E(\pi + \alpha - \beta)] \quad (158)$$

$$I_o = \frac{V_o - E}{R} = \frac{1}{\pi R} [V_m (1 + \cos \alpha) - E(\beta - \alpha)] \quad (159)$$

$$V_{or} = \sqrt{\frac{1}{\pi} \left[V_s^2 \left\{ (\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right\} + E^2(\pi + \alpha - \beta) \right]} \quad (160)$$

In all cases $FF = V_{or}/V_o$, $RF = \sqrt{FF^2 - 1}$, $I_{s,avg} = 0$ and $PIV_T = V_m$ (bridge).

8 Semi Converter

8.1 Operation with Inductive Load

In the semi converter the output voltage can never be negative. At $\omega t = \pi$ the load current is transferred to a free wheeling path inside the bridge itself. This is called inherent free wheeling.

- **Mode 1** ($\alpha \leq \omega t \leq \pi$): one thyristor and one diode conduct in diagonal. $v_o = v_s$, $i_s = i_o$
- **Mode 2** ($\pi < \omega t < \pi + \alpha$): free wheeling. $v_o = 0$, $i_s = 0$

Free wheeling path:

- **Symmetrical circuit:** the conducting thyristor and the diode of the same leg
- **Asymmetrical circuit:** the two diodes D1 and D2 in series
- **With separate FD:** the FD, since it offers one device drop against two

The output voltage waveform is therefore identical to that of the full converter with FD.

8.2 Output Expressions for Each Load

Semi converter load	Expressions are the same as
R (without or with FD)	Full converter with R load
RL (without or with FD)	Full converter with RL load with FD
RE (without or with FD)	Full converter with RE load
RLE (without or with FD)	Full converter with RLE load with FD

RL and RLE loads, continuous conduction:

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d(\omega t) = \frac{V_m}{\pi} (1 + \cos \alpha) \quad (161)$$

$$V_{or} = V_s \sqrt{\frac{1}{\pi} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]} \quad (162)$$

$$I_o = \frac{V_o}{R} \quad (\text{RL load}), \quad I_o = \frac{V_o - E}{R} \quad (\text{RLE load}) \quad (163)$$

$$\text{FF} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha}}{1 + \cos \alpha}, \quad \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (164)$$

V_o varies from $2V_m/\pi$ to zero as α varies from 0 to π . V_o cannot be negative, so inverter operation is not possible. The semi converter is a one quadrant converter and the full converter is a two quadrant converter.

8.3 Input Current (Ripple Free Load Current)

The source current is $+I_o$ for $(\pi - \alpha)$, zero for α and $-I_o$ for $(\pi - \alpha)$, for both arrangements:

$$I_{s,\text{avg}} = 0 \quad (165)$$

$$I_{s,\text{rms}} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} I_o^2 d(\omega t)} = I_o \sqrt{\frac{\pi - \alpha}{\pi}} \quad (166)$$

$$I_{s1} = \frac{2\sqrt{2}}{\pi} I_o \cos \frac{\alpha}{2}, \quad \text{DF} = \cos \frac{\alpha}{2} \quad (167)$$

$$\text{pf} = \frac{V_o I_o}{V_s I_{s,\text{rms}}} = \frac{\sqrt{2}(1 + \cos \alpha)}{\sqrt{\pi(\pi - \alpha)}} \quad (168)$$

The power factor of the semi converter is better than that of the full converter for the same α .

8.4 Device Currents (Ripple Free Load Current)

Arrangement	Thyristor conducts	Diode conducts	FD conducts
Symmetrical, no FD	π	π	not used
Asymmetrical, no FD	$\pi - \alpha$	$\pi + \alpha$	not used
Either, with FD	$\pi - \alpha$	$\pi - \alpha$	2α

The conduction angles are per supply cycle of 2π . For a device carrying I_o for an angle ϕ in 2π :

$$I_{\text{avg}} = I_o \frac{\phi}{2\pi}, \quad I_{\text{rms}} = I_o \sqrt{\frac{\phi}{2\pi}} \quad (169)$$

Symmetrical, no FD:

$$I_{T,\text{avg}} = I_{D,\text{avg}} = \frac{I_o}{2}, \quad I_{T,\text{rms}} = I_{D,\text{rms}} = \frac{I_o}{\sqrt{2}} \quad (170)$$

Asymmetrical, no FD:

$$I_{T,\text{avg}} = I_o \frac{\pi - \alpha}{2\pi}, \quad I_{D,\text{avg}} = I_o \frac{\pi + \alpha}{2\pi} \quad (171)$$

With FD:

$$I_{T,\text{avg}} = I_{D,\text{avg}} = I_o \frac{\pi - \alpha}{2\pi}, \quad I_{FD,\text{avg}} = I_o \frac{\alpha}{\pi} \quad (172)$$

8.5 PIV

$$\text{PIV}_T = \text{PIV}_D = \text{PIV}_{FD} = V_m \quad (173)$$

Example 8.1 (Semi Converter with Highly Inductive Load):

A 1-phase semi converter feeds a highly inductive load with $R = 5 \Omega$ from 230 V, 50 Hz supply at $\alpha = 45^\circ$.

Output voltage and current:

$$V_o = 103.54 (1 + \cos 45^\circ) = 103.54 \times 1.7071 = 176.75 \text{ V} \quad (174)$$

$$I_o = \frac{176.75}{5} = 35.35 \text{ A} \quad (175)$$

RMS output voltage, FF and RF:

$$V_{or} = 230 \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{4} + \frac{\sin 90^\circ}{2} \right)} = 230 \times 0.9535 = 219.30 \text{ V} \quad (176)$$

$$\text{FF} = \frac{219.30}{176.75} = 1.241, \quad \text{RF} = \sqrt{1.241^2 - 1} = 0.734 \quad (177)$$

Input current and power factor:

$$I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = 35.35 \sqrt{\frac{135}{180}} = 30.61 \text{ A} \quad (178)$$

$$\text{pf} = \frac{176.75 \times 35.35}{230 \times 30.61} = 0.887 \text{ (lag)} \quad (179)$$

Device currents:

$$\text{Symmetrical: } I_{T,\text{avg}} = I_{D,\text{avg}} = \frac{35.35}{2} = 17.68 \text{ A} \quad (180)$$

$$\text{Asymmetrical: } I_{T,\text{avg}} = 35.35 \times \frac{135}{360} = 13.26 \text{ A}, \quad I_{D,\text{avg}} = 35.35 \times \frac{225}{360} = 22.09 \text{ A} \quad (181)$$

$$\text{With FD: } I_{T,\text{avg}} = 13.26 \text{ A}, \quad I_{FD,\text{avg}} = 35.35 \times \frac{45}{180} = 8.84 \text{ A} \quad (182)$$

Comparison: the full converter gives $V_o = 146.42 \text{ V}$, $\text{RF} = 1.211$ and $\text{pf} = 0.637$ for the same data.

9 Effect of Source Inductance and Overlap Angle

Assumption: load current is constant at I_o .

9.1 Full Converter

Step 1: at $\omega t = \alpha$, T1 and T2 are fired while T3 and T4 are still conducting. All four thyristors conduct during overlap. The load is short circuited, so $v_o = 0$, and the source voltage appears across L_s :

$$L_s \frac{di_s}{dt} = V_m \sin \omega t \quad (183)$$

Step 2: the source current changes from $-I_o$ at $\omega t = \alpha$ to $+I_o$ at $\omega t = \alpha + \mu$:

$$\omega L_s \int_{-I_o}^{+I_o} di_s = \int_{\alpha}^{\alpha+\mu} V_m \sin \omega t d(\omega t) \quad (184)$$

$$2\omega L_s I_o = V_m [\cos \alpha - \cos(\alpha + \mu)] \quad (185)$$

$$\cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s I_o}{V_m} \quad (186)$$

Step 3: average output voltage. $v_o = 0$ from α to $\alpha + \mu$ and $v_o = v_s$ from $\alpha + \mu$ to $\pi + \alpha$:

$$V_o = \frac{1}{\pi} \int_{\alpha+\mu}^{\pi+\alpha} V_m \sin \omega t d(\omega t) = \frac{V_m}{\pi} [\cos(\alpha + \mu) + \cos \alpha] \quad (187)$$

$$= \frac{V_m}{\pi} \left[2 \cos \alpha - \frac{2\omega L_s I_o}{V_m} \right] \quad (188)$$

$$V_o = \frac{2V_m}{\pi} \cos \alpha - \frac{2\omega L_s}{\pi} I_o = \frac{2V_m}{\pi} \cos \alpha - 4fL_s I_o \quad (189)$$

Voltage drop due to overlap: $\Delta V_o = 4fL_s I_o$.

9.2 Semi Converter and Full Converter with FD

Step 1: at $\omega t = \alpha$ the load current is transferred from the free wheeling path to the source. The source current rises from 0 to I_o and $v_o = 0$ during this overlap:

$$\omega L_s I_o = V_m [\cos \alpha - \cos(\alpha + \mu)] \quad (190)$$

$$\cos(\alpha + \mu) = \cos \alpha - \frac{\omega L_s I_o}{V_m} \quad (191)$$

Step 2: at $\omega t = \pi$ the current returns to the free wheeling path. During this second overlap $v_o = 0$, which is the same as the ideal case. No further drop appears.

Step 3: average output voltage.

$$V_o = \frac{1}{\pi} \int_{\alpha+\mu}^{\pi} V_m \sin \omega t d(\omega t) = \frac{V_m}{\pi} [1 + \cos(\alpha + \mu)] \quad (192)$$

$$V_o = \frac{V_m}{\pi} (1 + \cos \alpha) - \frac{\omega L_s}{\pi} I_o = \frac{V_m}{\pi} (1 + \cos \alpha) - 2fL_s I_o \quad (193)$$

Voltage drop due to overlap: $\Delta V_o = 2fL_s I_o$.

Example 9.1 (Overlap Angle of Full Converter):

A 1-phase full converter feeds a level load current of 15 A from 230 V, 50 Hz supply. The source inductance is 2 mH and $\alpha = 30^\circ$.

Source reactance:

$$\omega L_s = 2\pi \times 50 \times 0.002 = 0.6283 \Omega \quad (194)$$

Overlap angle:

$$\cos(\alpha + \mu) = \cos 30^\circ - \frac{2 \times 0.6283 \times 15}{325.27} = 0.8660 - 0.0580 = 0.8081 \quad (195)$$

$$\alpha + \mu = 36.09^\circ \Rightarrow \mu = 6.09^\circ \quad (196)$$

Average output voltage:

$$V_o(\text{ideal}) = 207.07 \times 0.8660 = 179.33 \text{ V} \quad (197)$$

$$\Delta V_o = 4 \times 50 \times 0.002 \times 15 = 6.00 \text{ V} \quad (198)$$

$$V_o = 179.33 - 6.00 = 173.33 \text{ V} \quad (199)$$

10 Numerical Determination of Extinction Angle β

10.1 First Step: Check the Conduction Mode

β is required only for discontinuous conduction. The mode is therefore checked first.

Load (without FD)	Conduction is continuous if
RL	$\alpha \leq \theta$
RLE	$\frac{V_m}{Z} \sin(\theta - \alpha) \coth\left(\frac{\pi}{2 \tan \theta}\right) \geq \frac{E}{R}$

As a final check, the computed β must satisfy $\beta < \pi + \alpha$.

10.2 Cases with Direct Formula

Load	Extinction angle
R (without or with FD)	$\beta = \pi$
RE (without or with FD)	$\beta = \pi - \delta, \quad \delta = \sin^{-1}(E/V_m)$
RLE with FD or semi converter (discontinuous)	$\beta = \pi + \tan \theta \ln \left(1 + \frac{R I_\pi}{E}\right)$

10.3 Newton-Raphson Formulation for RL Load without FD

$$f(\beta) = \sin(\beta - \theta) - K e^{-\frac{\beta - \alpha}{\tan \theta}}, \quad K = \sin(\alpha - \theta) \quad (200)$$

$$f'(\beta) = \cos(\beta - \theta) + \frac{K}{\tan \theta} e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (201)$$

$$\beta_{k+1} = \beta_k - \frac{f(\beta_k)}{f'(\beta_k)}, \quad \beta_0 = \pi + \theta \quad (202)$$

Stopping rule: $|\beta_{k+1} - \beta_k| < 10^{-4}$ rad.

Caution: $\omega t = \alpha$ is also a root, since $f(\alpha) = 0$. It is the turn-on instant and must be rejected. The initial guess is taken near $\pi + \theta$. All angles must be in radian.

Fixed point alternative:

$$\beta_{k+1} = \pi + \theta - \sin^{-1} \left[K e^{-\frac{\beta_k - \alpha}{\tan \theta}} \right] \quad (203)$$

Example 10.1 (Full Converter with RL Load: β by NR Method):

A 1-phase full converter feeds an RL load with $R = 10 \, \Omega$ and $L = 31.83 \, \text{mH}$ from 230 V, 50 Hz supply at $\alpha = 60^\circ$. No FD is used and β is not given. Find β , γ , V_o and I_o .

Load data:

$$\omega L = 10 \, \Omega, \quad \theta = 45^\circ = 0.785398 \, \text{rad}, \quad \tan \theta = 1 \quad (204)$$

$$\alpha = 60^\circ = 1.047198 \, \text{rad} \quad (205)$$

Mode check: $\alpha = 60^\circ > \theta = 45^\circ$. Conduction is discontinuous and β is required.

Functions: $K = \sin(60^\circ - 45^\circ) = 0.258819$.

$$f(\beta) = \sin(\beta - 0.785398) - 0.258819 e^{-(\beta - 1.047198)} \quad (206)$$

$$f'(\beta) = \cos(\beta - 0.785398) + 0.258819 e^{-(\beta - 1.047198)} \quad (207)$$

Initial guess: $\beta_0 = \pi + \theta = 3.926991 \, \text{rad}$.

Iteration 1:

$$e^{-(3.926991 - 1.047198)} = e^{-2.879793} = 0.056147 \quad (208)$$

$$f(\beta_0) = \sin(\pi) - 0.258819 \times 0.056147 = -0.014532 \quad (209)$$

$$f'(\beta_0) = \cos(\pi) + 0.014532 = -0.985468 \quad (210)$$

$$\beta_1 = 3.926991 - \frac{-0.014532}{-0.985468} = 3.912245 \, \text{rad} \quad (211)$$

Iteration 2:

$$e^{-(3.912245 - 1.047198)} = e^{-2.865047} = 0.056981 \quad (212)$$

$$f(\beta_1) = \sin(3.126847) - 0.258819 \times 0.056981 = 0.014745 - 0.014748 = -0.000003 \quad (213)$$

$$f'(\beta_1) = -0.999891 + 0.014748 = -0.985143 \quad (214)$$

$$\beta_2 = 3.912245 - \frac{-0.000003}{-0.985143} = 3.912242 \, \text{rad} \quad (215)$$

Result:

$$\beta = 3.9122 \, \text{rad} = 224.16^\circ \quad (216)$$

$$\gamma = 224.16^\circ - 60^\circ = 164.16^\circ \quad (217)$$

Final check: $\beta = 224.16^\circ < \pi + \alpha = 240^\circ$. Conduction is discontinuous, as assumed. Since $\alpha > \theta$, β is slightly below $\pi + \theta = 225^\circ$.

Average output voltage and current:

$$V_o = \frac{325.27}{\pi} (\cos 60^\circ - \cos 224.16^\circ) = 103.54 (0.5 + 0.7174) = 126.05 \text{ V} \quad (218)$$

$$I_o = \frac{126.05}{10} = 12.60 \text{ A} \quad (219)$$

10.4 Newton-Raphson Formulation for RLE Load without FD

With $B = \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta)$, from (137):

$$g(\beta) = \frac{V_m}{Z} \sin(\beta - \theta) - \frac{E}{R} + B e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (220)$$

$$g'(\beta) = \frac{V_m}{Z} \cos(\beta - \theta) - \frac{B}{\tan \theta} e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (221)$$

$$\beta_{k+1} = \beta_k - \frac{g(\beta_k)}{g'(\beta_k)} \quad (222)$$

Initial guess: neglecting the exponential term gives $\frac{V_m}{Z} \sin(\beta - \theta) = \frac{E}{R}$. Since $\frac{Z}{R} = \frac{1}{\cos \theta}$:

$$\sin(\beta - \theta) = \frac{E}{R} \cdot \frac{Z}{V_m} = \frac{E}{V_m \cos \theta} \quad (223)$$

Taking the root on the falling part of the sine wave:

$$\beta_0 = \pi + \theta - \sin^{-1} \left(\frac{E}{V_m \cos \theta} \right) \quad (224)$$

If $E > V_m \cos \theta$, $\beta_0 = \pi$ is used.

Example 10.2 (Full Converter with RLE Load: β by NR Method):

A 1-phase full converter feeds an RLE load with $R = 10 \Omega$, $\omega L = 10 \Omega$ and $E = 100 \text{ V}$ from 230 V, 50 Hz supply at $\alpha = 45^\circ$. No FD is used and β is not given. Find β , γ , I_o and V_o .

Load data:

$$\theta = 45^\circ = 0.785398 \text{ rad}, \quad \tan \theta = 1, \quad Z = 14.142 \Omega \quad (225)$$

$$\frac{V_m}{Z} = 23 \text{ A}, \quad \frac{E}{R} = 10 \text{ A} \quad (226)$$

$$\delta = \sin^{-1} \left(\frac{100}{325.27} \right) = 17.90^\circ < \alpha \quad \checkmark \quad (227)$$

Mode check:

$$\frac{V_m}{Z} \sin(\theta - \alpha) \coth \left(\frac{\pi}{2 \tan \theta} \right) = 0 < \frac{E}{R} = 10 \text{ A} \quad (228)$$

Conduction is discontinuous and β is required.

Functions: $B = 10 - 23 \sin 0^\circ = 10 \text{ A}$. Let $u = \beta - \theta$. In this example $\alpha = \theta$, so $\beta - \alpha$ is also equal to u .

$$g(u) = 23 \sin u - 10 + 10 e^{-u} \quad (229)$$

$$g'(u) = 23 \cos u - 10 e^{-u} \quad (230)$$

Initial guess:

$$u_0 = \pi - \sin^{-1} \left(\frac{10}{23} \right) = 3.14159 - 0.44979 = 2.69180 \text{ rad} \quad (231)$$

Iteration 1:

$$g(u_0) = 23 \times 0.43478 - 10 + 10 \times 0.067759 = 0.67759 \quad (232)$$

$$g'(u_0) = 23 \times (-0.90054) - 0.67759 = -21.3900 \quad (233)$$

$$u_1 = 2.69180 - \frac{0.67759}{-21.3900} = 2.72348 \text{ rad} \quad (234)$$

Iteration 2:

$$g(u_1) = 23 \times 0.406037 - 10 + 10 \times 0.065646 = -0.00468 \quad (235)$$

$$g'(u_1) = 23 \times (-0.913857) - 0.65646 = -21.6752 \quad (236)$$

$$u_2 = 2.72348 - \frac{-0.00468}{-21.6752} = 2.72326 \text{ rad} \quad (237)$$

Iteration 3: $|u_2 - u_1| = 0.00022 \text{ rad}$ and $g(u_2) \approx 0$. The solution has converged.

Result:

$$\beta = u + \theta = 2.7233 + 0.7854 = 3.5087 \text{ rad} = 201.03^\circ \quad (238)$$

$$\gamma = 201.03^\circ - 45^\circ = 156.03^\circ = 2.7233 \text{ rad} \quad (239)$$

Final check: $\beta = 201.03^\circ < \pi + \alpha = 225^\circ$. Conduction is discontinuous, as assumed.

Average output current and voltage:

$$I_o = \frac{1}{\pi \times 10} [325.27 (\cos 45^\circ - \cos 201.03^\circ) - 100 \times 2.7233] \quad (240)$$

$$= \frac{325.27 \times 1.6405 - 272.33}{31.416} = \frac{533.60 - 272.33}{31.416} = 8.32 \text{ A} \quad (241)$$

$$V_o = E + R I_o = 100 + 10 \times 8.32 = 183.2 \text{ V} \quad (242)$$

Remark: $u = 0$ (i.e. $\beta = \alpha$) also satisfies $g(u) = 0$. It is the turn-on instant and must be rejected.

11 Summary of Derived Expressions

11.1 Average Output Voltage and Current

Table 2: Average output voltage and average output current

Converter and load	V_o	I_o
Full or semi, R load	$\frac{V_m}{\pi} (1 + \cos \alpha)$	$\frac{V_o}{R}$
Full, RL, no FD, discontinuous	$\frac{V_m}{\pi} (\cos \alpha - \cos \beta)$	$\frac{V_o}{R}$

Table 2 (continued from previous page)

Converter and load	V_o	I_o
Full, RL, no FD, continuous	$\frac{2V_m}{\pi} \cos \alpha$	$\frac{V_o}{R}$
Full with FD or semi, RL	$\frac{V_m}{\pi} (1 + \cos \alpha)$	$\frac{V_o}{R}$
Full or semi, RE load	$\frac{1}{\pi} [V_m (\cos \alpha + \cos \delta) + E(\alpha + \delta)]$	$\frac{V_o - E}{R}$
Full, RLE, no FD, discontinuous	$\frac{1}{\pi} [V_m (\cos \alpha - \cos \beta) + E(\pi + \alpha - \beta)]$	$\frac{V_o - E}{R}$
Full, RLE, no FD, continuous	$\frac{2V_m}{\pi} \cos \alpha$	$\frac{V_o - E}{R}$
Full with FD or semi, RLE, continuous	$\frac{V_m}{\pi} (1 + \cos \alpha)$	$\frac{V_o - E}{R}$
Full with FD or semi, RLE, discontinuous	$\frac{1}{\pi} [V_m (1 + \cos \alpha) + E(\pi + \alpha - \beta)]$	$\frac{V_o - E}{R}$

11.2 RMS Output Voltage

Table 3: RMS output voltage

Converter and load	V_{or}^2
Full or semi, R load	$\frac{V_s^2}{\pi} [(\pi - \alpha) + \frac{1}{2} \sin 2\alpha]$
Full, RL, no FD, discontinuous	$\frac{V_s^2}{\pi} [\gamma - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)]$
Full, RL or RLE, no FD, continuous	V_s^2
Full with FD or semi, RL or RLE, continuous	$\frac{V_s^2}{\pi} [(\pi - \alpha) + \frac{1}{2} \sin 2\alpha]$
Full or semi, RE load	$\frac{1}{\pi} [V_s^2 \{ \gamma + \frac{1}{2}(\sin 2\alpha + \sin 2\delta) \} + E^2(\alpha + \delta)]$
Full, RLE, no FD, discontinuous	$\frac{1}{\pi} [V_s^2 \{ \gamma - \frac{1}{2}(\sin 2\beta - \sin 2\alpha) \} + E^2(\pi - \gamma)]$
Full with FD or semi, RLE, discontinuous	$\frac{1}{\pi} [V_s^2 \{ (\pi - \alpha) + \frac{1}{2} \sin 2\alpha \} + E^2(\pi + \alpha - \beta)]$

11.3 Input Current, PIV, FF and RF

- All full wave cases: $I_{s,avg} = 0$, $FF = \frac{V_{or}}{V_o}$ and $RF = \sqrt{FF^2 - 1}$

- **Full converter without FD:** $I_{s,\text{rms}} = I_o$, $I_{T,\text{avg}} = \frac{I_o}{2}$, $I_{T,\text{rms}} = \frac{I_o}{\sqrt{2}}$
- **Full converter, level current:** $I_{s,\text{rms}} = I_o$ and $\text{pf} = \frac{2\sqrt{2}}{\pi} \cos \alpha$
- **Semi converter or full converter with FD, level current:** $I_{s,\text{rms}} = I_o \sqrt{\frac{\pi - \alpha}{\pi}}$ and $\text{pf} = \frac{\sqrt{2}(1 + \cos \alpha)}{\sqrt{\pi(\pi - \alpha)}}$
- **PIV:** V_m for bridge circuits and $2V_m$ for mid-point circuit
- **Overlap:** $\Delta V_o = 4fL_sI_o$ for full converter and $2fL_sI_o$ for semi converter

11.4 Comparison of Converters at $\alpha = 0$ with R Load

Quantity	Half wave	Full wave (full or semi)
V_o	V_m/π	$2V_m/\pi$
V_{or}	$V_m/2$	$V_m/\sqrt{2}$
FF	1.571	1.111
RF	1.211	0.483
η	40.53%	81.06%
Input pf	0.707	1.0
Ripple frequency	f	$2f$
DC component in i_s	present	absent

12 Practice Problems

12.1 Full Converter with R Load

Problem 12.1 (R Load at $\alpha = 90^\circ$):

A 1-phase full converter feeds $R = 20 \Omega$ from 230 V, 50 Hz supply. For $\alpha = 90^\circ$ determine V_o , V_{or} , I_o , I_{or} , FF and RF.

Solution:

Average values:

$$V_o = \frac{325.27}{\pi} (1 + \cos 90^\circ) = 103.54 \text{ V} \quad (243)$$

$$I_o = \frac{103.54}{20} = 5.18 \text{ A} \quad (244)$$

RMS values:

$$V_{or} = 230 \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{2} + \frac{\sin 180^\circ}{2} \right)} = 230 \times 0.7071 = 162.63 \text{ V} \quad (245)$$

$$I_{or} = \frac{162.63}{20} = 8.13 \text{ A} \quad (246)$$

Form factor and ripple factor:

$$FF = \frac{162.63}{103.54} = 1.571 \quad (247)$$

$$RF = \sqrt{1.571^2 - 1} = 1.211 \quad (248)$$

Conclusion: FF rises from 1.111 at $\alpha = 0$ to 1.571 at $\alpha = 90^\circ$.

12.2 Full Converter with RLE Load

Problem 12.2 (Rectifier and Inverter Operation):

A 1-phase full converter feeds an RLE load with $R = 2 \Omega$ and very large L from 230 V, 50 Hz supply. (a) For $E = 40$ V and $\alpha = 30^\circ$ find V_o , I_o and input pf. (b) For $E = -150$ V and $\alpha = 120^\circ$ find V_o , I_o and the power fed back to the source.

Solution:

(a) Rectifier operation:

$$V_o = \frac{2 \times 325.27}{\pi} \cos 30^\circ = 207.07 \times 0.8660 = 179.33 \text{ V} \quad (249)$$

$$I_o = \frac{179.33 - 40}{2} = 69.67 \text{ A} \quad (250)$$

$$\text{pf} = 0.9 \cos 30^\circ = 0.780 \text{ (lag)} \quad (251)$$

(b) Inverter operation:

$$V_o = 207.07 \times \cos 120^\circ = -103.54 \text{ V} \quad (252)$$

$$I_o = \frac{-103.54 - (-150)}{2} = 23.23 \text{ A} \quad (253)$$

Power balance:

$$\text{Power supplied by } E = 150 \times 23.23 = 3484.5 \text{ W} \quad (254)$$

$$\text{Loss in } R = 23.23^2 \times 2 = 1079.3 \text{ W} \quad (255)$$

$$\text{Power fed back to source} = 103.54 \times 23.23 = 2405.2 \text{ W} \quad (256)$$

Conclusion: $3484.5 = 1079.3 + 2405.2$ W. The power balance is satisfied.

12.3 Semi Converter with RLE Load

Problem 12.3 (Semi Converter, Continuous Conduction):

A 1-phase semi converter feeds an RLE load with $R = 2 \Omega$, $E = 40$ V and very large L from 230 V, 50 Hz supply at $\alpha = 60^\circ$. Determine V_o , I_o , $I_{s,\text{rms}}$ and input pf.

Solution:

Average output voltage and current:

$$V_o = 103.54 (1 + \cos 60^\circ) = 155.31 \text{ V} \quad (257)$$

$$I_o = \frac{155.31 - 40}{2} = 57.65 \text{ A} \quad (258)$$

RMS input current:

$$I_{s,\text{rms}} = 57.65 \sqrt{\frac{120}{180}} = 57.65 \times 0.8165 = 47.07 \text{ A} \quad (259)$$

Input power factor:

$$\text{pf} = \frac{155.31 \times 57.65}{230 \times 47.07} = 0.827 \text{ (lag)} \quad (260)$$

Check by formula:

$$\text{pf} = \frac{\sqrt{2}(1 + \cos 60^\circ)}{\sqrt{\pi(\pi - \pi/3)}} = \frac{2.1213}{2.5651} = 0.827 \quad (261)$$

Conclusion: the full converter at the same α gives $\text{pf} = 0.9 \cos 60^\circ = 0.45$ only.

13 Summary and Key Learning Outcomes

13.1 Fundamental Concepts Mastered

Through the present note, students have developed thorough understanding of:

1. **Two pulse operation:** output period π , double average voltage and lower ripple than the half wave circuit
2. **Input current:** zero average value, so no DC magnetisation of the supply transformer
3. **Conduction modes:** continuous conduction for $\alpha \leq \theta$ (RL load) and discontinuous conduction otherwise
4. **Full converter:** $V_o = (2V_m/\pi) \cos \alpha$, two quadrant operation and inversion for $\alpha > \pi/2$
5. **Semi converter:** $V_o = (V_m/\pi)(1 + \cos \alpha)$, inherent free wheeling, one quadrant operation and better power factor
6. **Effect of source inductance:** voltage drop $4fL_sI_o$ for full converter and $2fL_sI_o$ for semi converter

13.2 Key Mathematical Relationships

$$\text{R load: } V_o = \frac{V_m}{\pi}(1 + \cos \alpha) \quad (262)$$

$$\text{Full converter, continuous: } V_o = \frac{2V_m}{\pi} \cos \alpha \quad (263)$$

$$\text{Semi converter, continuous: } V_o = \frac{V_m}{\pi}(1 + \cos \alpha) \quad (264)$$

$$\text{RE load: } I_o = \frac{1}{\pi R} [V_m(\cos \alpha + \cos \delta) - E\gamma] \quad (265)$$

$$\text{Ripple factor: } \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (266)$$

$$\text{Overlap (full converter): } \cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s I_o}{V_m} \quad (267)$$

Definition 13.1 (Module Integration): Every full wave expression is obtained from the corresponding half wave expression by changing the averaging period from 2π to π . The current expressions and the extinction angle equations remain the same. The same method is extended to three-phase converters in the subsequent lectures.

14 Nomenclature and Abbreviations

14.1 Symbols

Symbol	Description
v_s	Instantaneous source voltage (V)
V_s	RMS source voltage (V)
V_m	Peak source voltage, $\sqrt{2}V_s$ (V)
v_o, i_o	Instantaneous output (load) voltage (V) and current (A)
V_o, I_o	Average output voltage (V) and current (A)
V_{or}, I_{or}	RMS output voltage (V) and current (A)
V_{ac}	RMS value of ripple component of output voltage (V)
ΔV_o	Drop in average output voltage due to overlap (V)
i_s	Instantaneous source (input) current (A)
$I_{s,avg}, I_{s,rms}$	Average and RMS input current (A)
I_{s1}	RMS value of fundamental component of input current (A)
i_T	Instantaneous thyristor current (A)
$I_{T,avg}, I_{T,rms}$	Average and RMS thyristor current (A)
$I_{D,avg}, I_{D,rms}$	Average and RMS current of bridge diode (A)
$I_{FD,avg}, I_{FD,rms}$	Average and RMS free wheeling diode current (A)
I_α, I_π	Load current at $\omega t = \alpha$ and $\omega t = \pi$ (A)
R	Load resistance (Ω)
L	Load inductance (H)
E	Load EMF or battery voltage (V)
L_s	Source inductance (H)
Z	Load impedance, $\sqrt{R^2 + (\omega L)^2}$ (Ω)
ω	Angular frequency of supply, $2\pi f$ (rad/s)
f	Supply frequency (Hz)
α	Firing angle (rad)
β	Extinction angle (rad)
γ	Conduction angle, $\beta - \alpha$ (rad)
θ	Load impedance angle, $\tan^{-1}(\omega L/R)$ (rad)
μ	Overlap angle (rad)
δ	Angle at which $v_s = E$, $\sin^{-1}(E/V_m)$ (rad)
ϕ	Conduction angle of a device per supply cycle (rad)
A, K, B	Constants of integration and substitution
J_1, J_2, J_3	Intermediate integrals of RMS current derivation
$f(\beta), g(\beta)$	Extinction functions of RL and RLE loads for NR method
β_k, u_k	Value of β and of $u = \beta - \theta$ at k -th iteration (rad)
P, P_E, P_R	Load power, power to E , loss in R (W)
η	Rectification efficiency
t_c	Circuit turn-off time (s)

14.2 Abbreviations

Abbreviation	Full Form
AC	Alternating Current
DC	Direct Current
DF	Displacement Factor
EMF	Electromotive Force
FD	Free Wheeling Diode
FF	Form Factor
KVL	Kirchhoff's Voltage Law
NR	Newton-Raphson
pf	Power Factor
PIV	Peak Inverse Voltage
R	Resistive Load
RE	Resistive Load with EMF
RF	Ripple Factor
RL	Resistive Inductive Load
RLE	Resistive Inductive Load with EMF
RMS	Root Mean Square
SCR	Silicon Controlled Rectifier
T	Thyristor
THD	Total Harmonic Distortion

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End of Class Notes: 1-Phase Full Wave Fully Controlled and Semi Controlled Rectifier

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